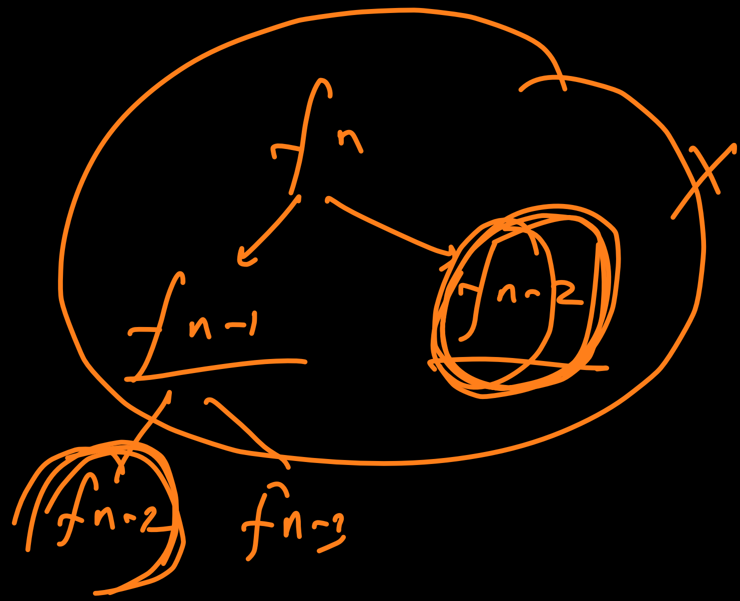
Greedy

↳ algorithmic Paradigms

- Brute force
- Dynamic Prog.
- Divide N Conquer → Merge Sort, quick, B.S
- Greedy



[2, 0, 5, -1, 2]

K = 4

greedy

local optimum

4, 10

(Maximize the sum)

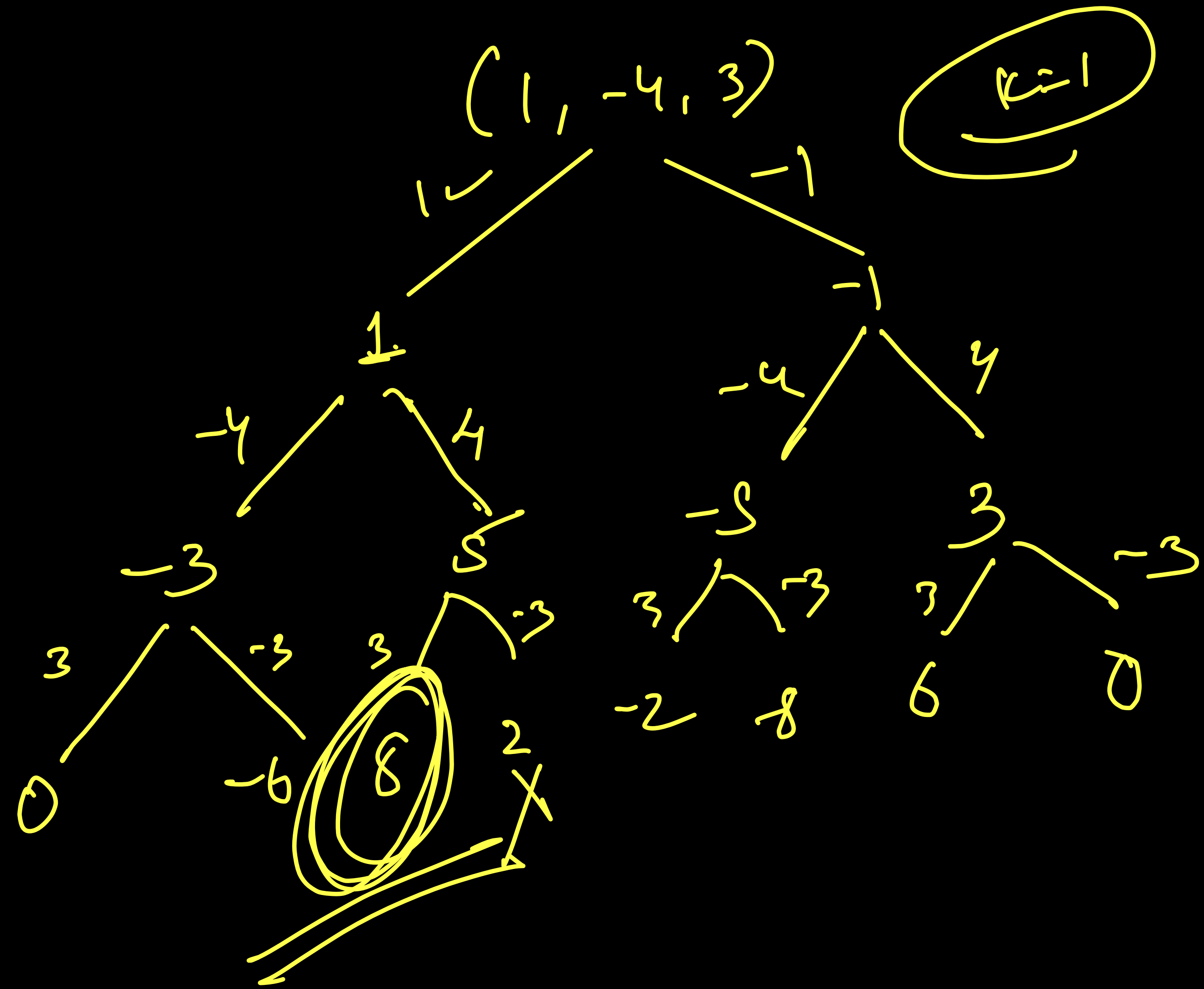
2 -2

Brute force

-1

X

In bruteforce you try to create all possibilities of the given no. such that once the no can be negated once it cannot.


(1, -4, 3)
k=1
1
-1
1
-1
-4
4
-4
4
-3
5
-5
3
3
-3
3
-3
3
-3
3
-3
0
-6
8
2
-2
-8
6
0

[5, 3, 2, 1, ~~1~~] 0 K = ~~3~~ ~~2~~ 1

9

Min heap

Smallest element

+ve → +

-ve →

$\boxed{\text{Sum - max}}$ < Sum - min

4

3, 4, 10, +α

$K = \cancel{3}\ \cancel{2}\ 1$

test cases

greedy

smart

intelli

matter

50-60% → Sorting → custom comp

→ pq

fractional knapsack → Bagpack

N items → break

item → weight → profit → Maximum Profit

knapsack → max capacity weight W.

1kg sugar → 100

1kg rice → 150

1kg wheat → 170

W → 2.3 kg

A1
51 + 100 + 150

A2
170 + 150 + 30

$A_2 > A_1$

→ N = 3   W = 50

item 1 → P → 60   W → 10

item 2   P → 100   W − 20

item 3   P → 120   W − 30

sort   → n log n

60 + 100 + 80

→ 100 + 140

→ 290

220

item → $\frac{\text{value}}{\text{weight}}$ → des

$i \to 0 \; W$

$\frac{100}{4} \quad \frac{300}{3} \quad \frac{200}{3}$

$\frac{\cancel{100}}{\cancel{4}}\ 25 \qquad 100 \qquad 66.6$

$\frac{500}{9}$ ↳ 9kg

$7 \times \frac{500}{9} \to 55.5$

↳ 388.8

$\frac{500}{9} \times 4 \to \frac{2000}{9}$

222.2

$\overset{0^{th}}{[1,3]}$, $\overset{1^{st}}{[2,2]}$, $\overset{2^{nd}}{[3,1]}$]

Size = ~~4~~ ~~3~~ ~~1~~ 0

Maximize profit

profit

0^{th} → no. of boxes → 1

↳ profit of each → 3

1^{st} → no. of boxes → 2 ✓

↳ profit of each → 2

2^{nd} → no. of boxes → 3

↳ profit of each → 1

3 + 4 + 1

~~8~~

JOIN THE DARKSIDE

boat $\rightarrow$ it can carry a max weight of $\rightarrow$ limit

$\rightarrow$ it can carry atmost 2 people together

$$people[i] \le limit$$

$\hookrightarrow$ there won't be any person whose weight more than boat limit

→ worst possible → n

→ Best ans → $\frac{n}{2}$

S 10 12

P_1 P_2 P_3 P_4 P_5

2 3 7 8 10 Dcut → 10

j i

→ sol

dcut → 12

B + ~~[illegible]~~ $O(\log n)$

Minimum Product Subset

array → { -ve, +ve, zero } → 1 → 10^5

Subset whose product is minimum. Return the product.

Ex → [-1, -1, -2, 4, 3]

↳ -24

Ex → [-5, 0]

↓

-5

Ex → [-5, 0, -4, -3]

→ -60

Brute force

↓

generate all possible subsets

$n \to \boxed{2^n}$

JOIN THE DARKSIDE

No negatives

[1, 0, 2, 3]

- no zeroes & +ve
  - ↓ min +ve elmt
- some zeroes & +ve
  - ↓ zero

:- If we have negatives

even

product of all -ve except the largest -ve.

[-1, 3, -2, -4, 8, -3]

odd

product of all -ve × product of all +ve

[1, -1, 3, -6, -2]

→ $-1 \times -6 \times -2 \rightarrow -12$

→ 3   -36

$-a \times -5 \rightarrow$ +ve

$-b \times -c \times -a \Rightarrow$ -ve

+ve

$-20 \quad x \quad 1 \rightarrow -20$

$2 \rightarrow -40$

x

$-2 \times -3 \quad x -4 \rightarrow 12$

x

6

— lexico → layout ⟶ **layer clue**

⟶ repeat limit ⟵ K = 3

⟶ ≤ given no. of clue

ch1 > ch2

ch1 ch1 ch1 . . . ch1

$\underbrace{\text{ch1 ch1 ch1 . . . ch1}}_{\text{K times}}$

bbb a bbbb

k = 3

z^1

array → desc

→ bbb a bbb

2nd largest

↳ 1st
2nd
3rd
⋮

max heap
↓
<char, freq>

aababab   k = 2

$\boxed{bbabaa}$